Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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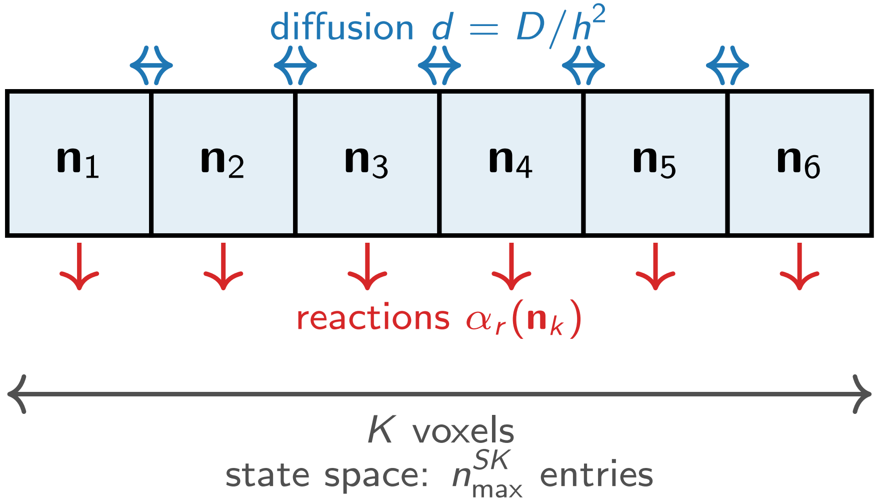
The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}^S_{\geq 0}$.

Two types of events:

- **Reactions** inside voxel k: rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ **(RDME)**.



The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in** K. Even $K{=}8$, $S{=}1$ is intractable with a naive FSP.

The probability ODE $\dot{p} = Ap$ has block-tridiagonal structure (schematic, $K{=}4$, $S{=}1$)

$$
\begin{array}{c|cccc}
 & \mathbf{n}_1 & \mathbf{n}_2 & \mathbf{n}_3 & \mathbf{n}_4 \\
\hline
\begin{array}{c} \mathbf{n}_1 \\ \mathbf{n}_2 \\ \mathbf{n}_3 \\ \mathbf{n}_4 \end{array} & \multicolumn{4}{c}{\begin{bmatrix} A_1 & D_{12} & & \\ D_{12} & A_2 & D_{23} & \\ & D_{23} & A_3 & D_{34} \\ & & D_{34} & A_4 \end{bmatrix}}
\end{array}
\cdot
\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix}
=
\begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{p}_4 \end{bmatrix}
$$

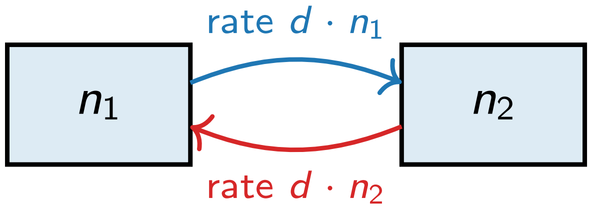
each block: $n_{\max}^{S} \times n_{\max}^{S}$ total: $n_{\max}^{SK} \times n_{\max}^{SK}$

A_k: reactions inside voxel k $D_{k,k+1}$: diffusion between neighboring voxels p_k: probability sub-vector for voxel k's state

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

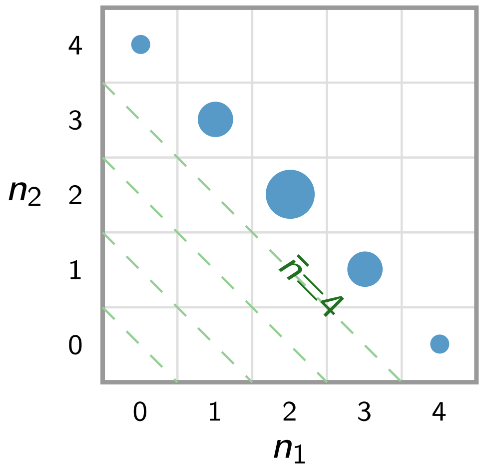
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently, so at equilibrium it is equally likely to be in either voxel. For $\bar{n}$ total:

$$\boxed{P(n_1 \mid \bar{n}) = \mathrm{Bin}\big(\bar{n},\ \tfrac{1}{2}\big)} \qquad \text{(exact, any } d > 0\text{)}$$

When the distribution is near this equilibrium (checked at runtime via α_j), $\bar{n}$ **alone describes the two-voxel state**.



Probability is spread along each constant-total diagonal as $\mathrm{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

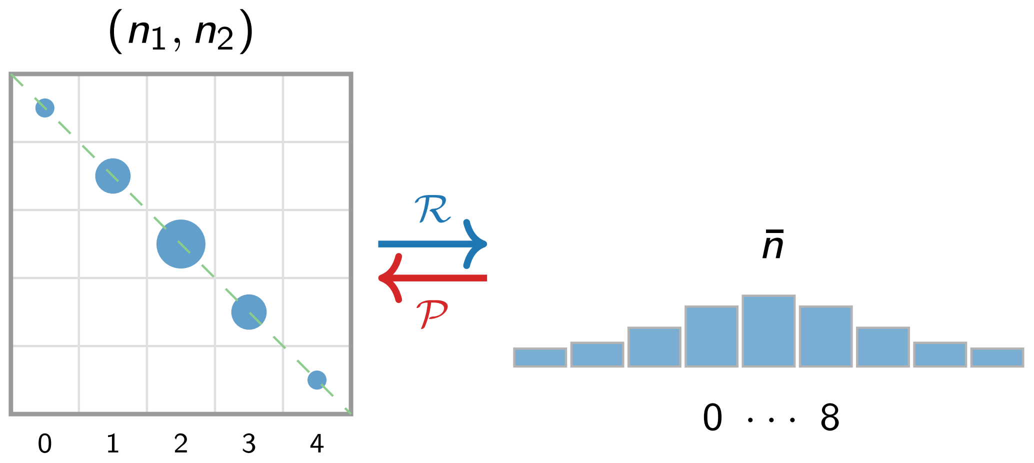
Merging two adjacent voxels into their total

Compress ($\mathcal{R}$) — exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$) — approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \mathrm{Bin}(\bar{n}, \tfrac{1}{2}).$$



Summing along constant-total diagonals (left) gives the 1-D total $\bar{n}$ (right). Reconstruction spreads mass back via Binomial.

State-space reduction

Per block: $n_{\max}^{2S} \to n_{\max}^{S}$. All K voxels: $n_{\max}^{SK} \to n_{\max}^{SK/2}$.

Reconstruct is valid when the distribution is near-Binomial — verified at runtime via α_j, not assumed globally.

What merging does to the rate matrix

Full rate matrix A — 4 voxels:

block 1 block 2

$$\begin{array}{c|cccc|} & n_1 & n_2 & n_3 & n_4 \\ \hline n_1 & A_1 & D_{12} & & \\ n_2 & D_{12} & A_2 & D_{23} & \\ n_3 & & D_{23} & A_3 & D_{34} \\ n_4 & & & D_{34} & A_4 \\ \hline \end{array}$$

Size: $n_{\max}^4$

$$\xrightarrow{\mathcal{R}} \qquad \xleftarrow{\mathcal{P}}$$

Merged rate matrix $\bar{A}$ — 2 blocks:

$$\begin{array}{c|cc|} & \bar{n}_1 & \bar{n}_2 \\ \hline \bar{n}_1 & \bar{A}_1 & \bar{D}_{12} \\ \bar{n}_2 & \bar{D}_{12} & \bar{A}_2 \\ \hline \end{array}$$

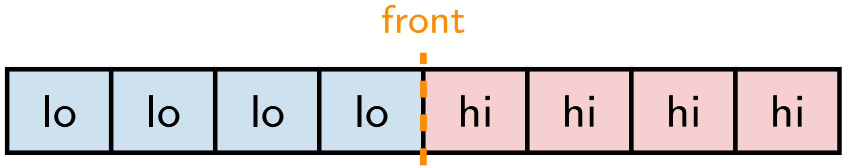
Size: $n_{\max}^2$ ($\mathbf{10^4}\times$ smaller for $n_{\max}=100$)

- $A_k \to \bar{A}_j$: reactions averaged over the Binomial
- Within-block D disappears — implicit in the Binomial
- $\bar{D}_{12}$: only between-block diffusion remains

Merging fails at a bistable front

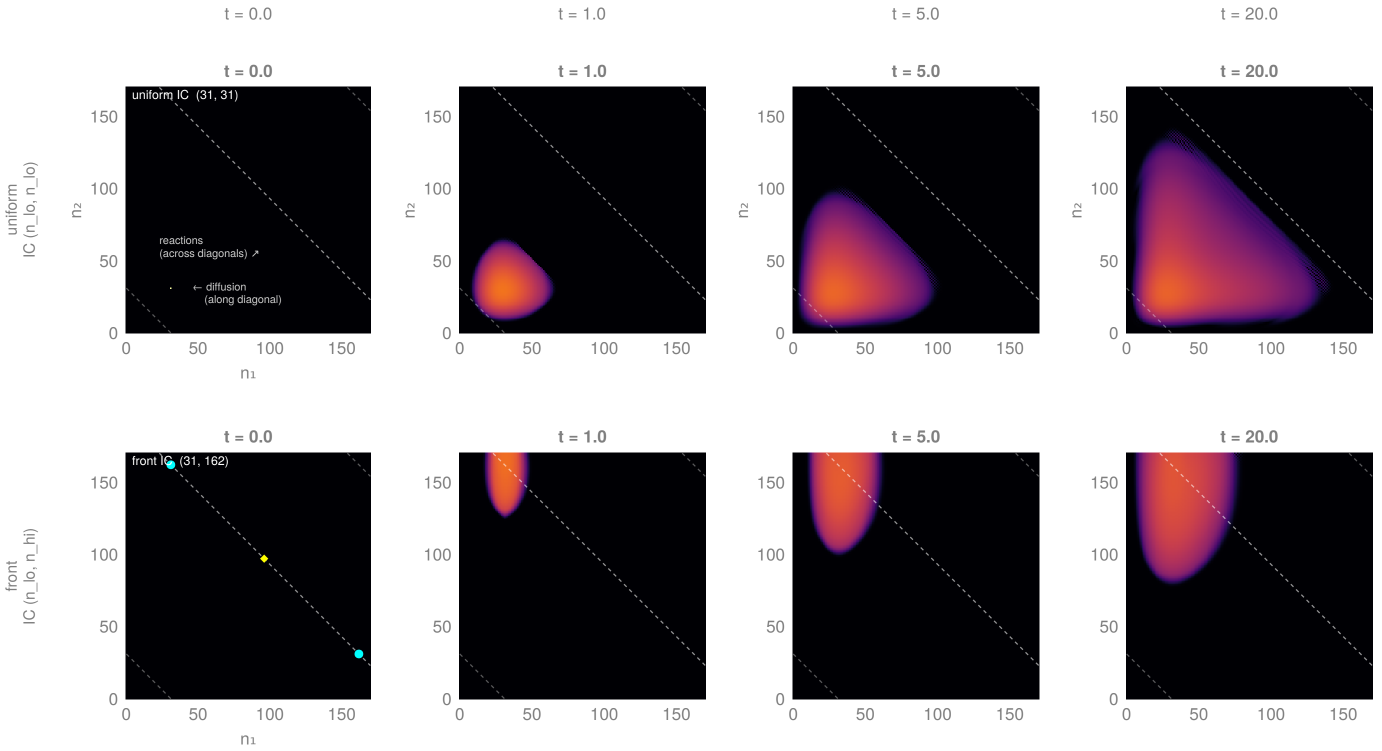
Schlögl model ($K=8$, $S=1$): each voxel has two stable states.

A **front** forms when neighboring voxels occupy opposite stable states. Slow diffusion ($d \ll$ reaction rates) sustains it indefinitely.



At the front: the joint distribution of the two interface voxels is *bimodal* — concentrated at $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$.

The Binomial is **unimodal**: wrong shape.



Top: uniform initial condition — probability spreads along constant-total diagonals; Binomial reconstruction is accurate. ✓

Bottom: bistable front — probability stays at the corners; Binomial reconstruction gives the wrong shape. ✗

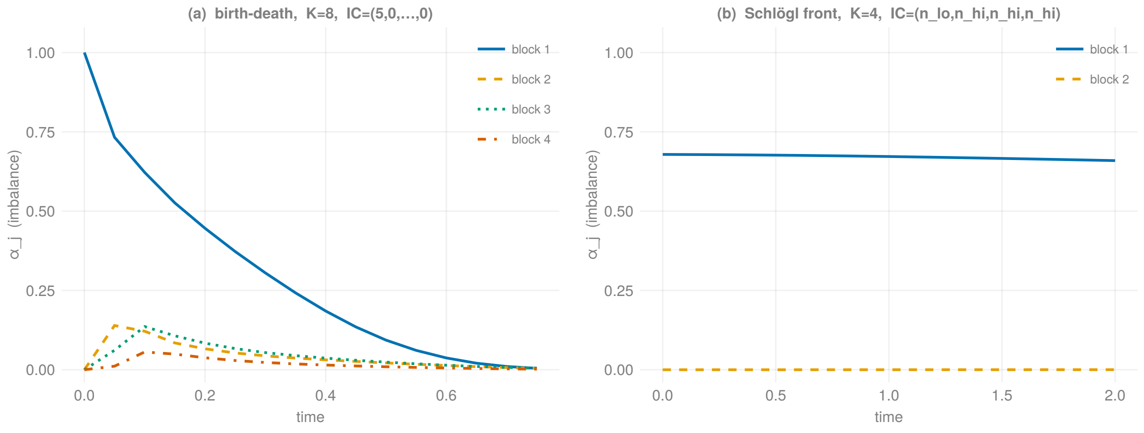
Compare mean counts in adjacent voxels $2j-1$ and $2j$:

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}.$$

- Small α_j: voxels balanced $\Rightarrow$ distribution near-Binomial $\Rightarrow$ safe to merge.
- Large α_j: voxels in opposite states $\Rightarrow$ front is nearby $\Rightarrow$ keep fine.

Merging criterion

Find the most imbalanced block:
$j^* = \mathrm{argmax}_j\ \alpha_j$.
Keep j^* and its immediate neighbors fine; merge all others.



(a) Birth-death: α_1 decays to zero as diffusion equilibrates the two voxels.
(b) Schlögl front: α_1 stays near 1 — the front is persistent.

Green band = safe region for merging.

Adaptive algorithm: mixed-resolution evolution

Idea. Keep a fine window at the front; coarsen everything else. Evolve the mixed state *directly* — no reconstruction step.

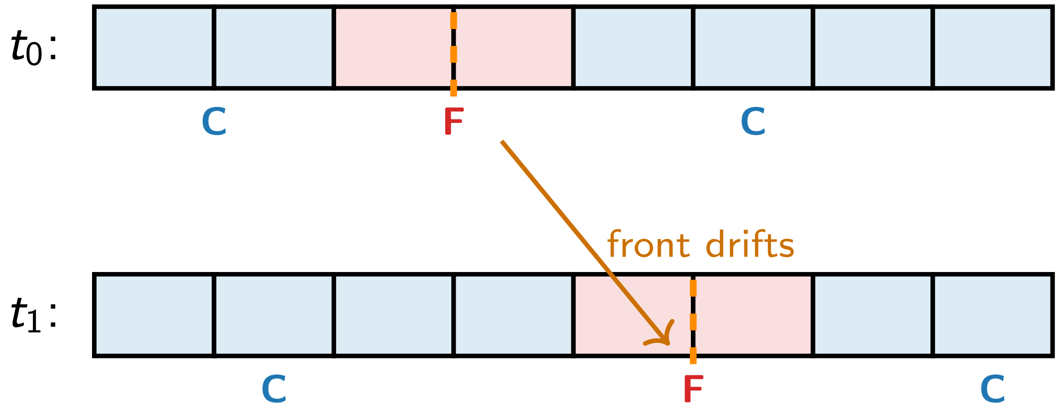
Mixed state p^M:

- **Coarse groups**: only the total $\bar{n}_j$ is tracked; split assumed Binomial.
- **Fine group at front**: both voxel counts (n_{2j-1}, n_{2j}) tracked exactly.

Evolution: $\dot{p}^M = A^M p^M$, where A^M couples merged and fine blocks at block boundaries.

Reassignment: recompute α_j only when the front crosses a block boundary; update the fine window to be centered on $\text{argmax}_j\ \alpha_j$.

Scope: one dominant, monotone-drifting front.



Fine window follows the front. Coarse regions stay compressed.

One time step of the adaptive solver

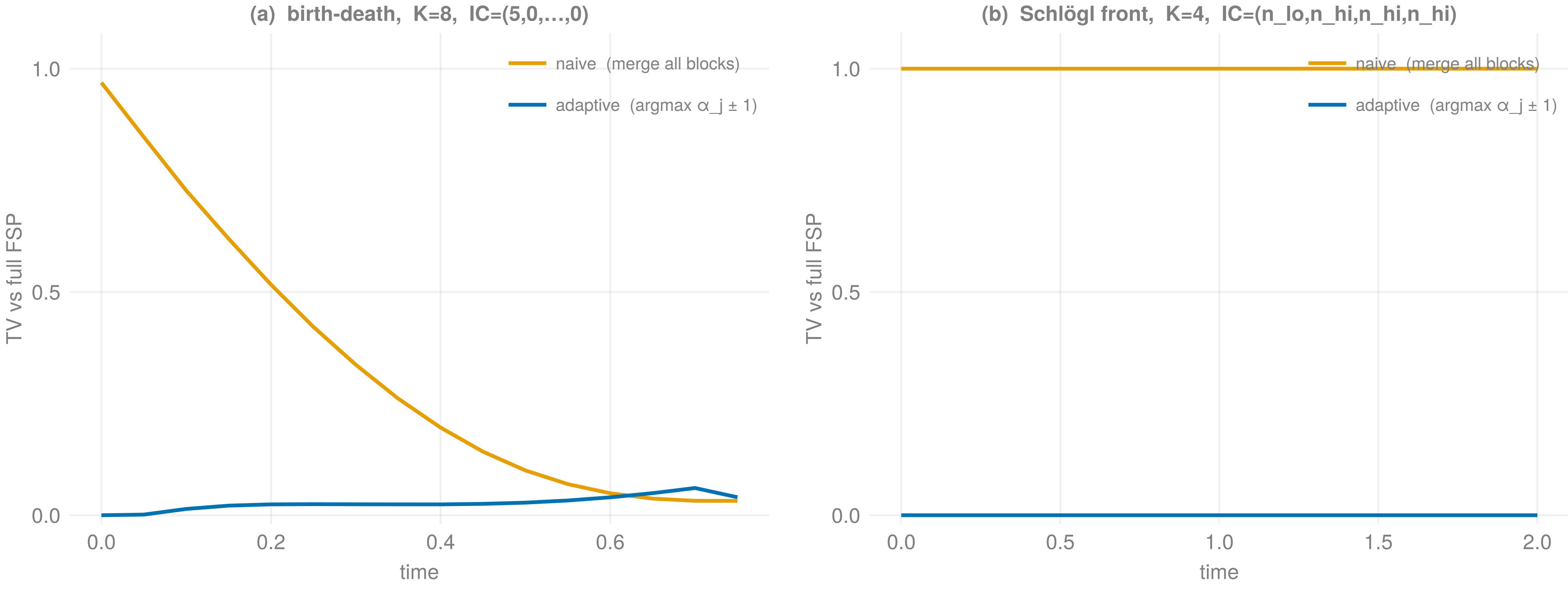
Every step:

1. **Solve:** $p^M \leftarrow e^{\Delta t\, A^M}\, p^M$
2. **Truncate/expand:** drop states below threshold τ; add one-step reachable states (adaptive FSP)
3. **Track front:** compute weighted mean block position $\bar{\jmath}$ from voxel means — $O(|\mathcal{S}^M|)$, cheap

Only when front crosses a block boundary ($|\bar{\jmath} - \bar{\jmath}_{\mathrm{prev}}| \geq \frac{1}{2}$):

4. **Re-evaluate** α_j for all blocks; find $j^* = \mathrm{argmax}_j\ \alpha_j$
5. **Update partition:** keep j^* and its neighbors fine; merge all other blocks
6. **Transfer state:** split blocks $\rightarrow$ fine via $\mathrm{Bin}(\bar{n}_j, \frac{1}{2})$; merged blocks $\rightarrow$ compressed via $\bar{n}_j = n_{2j-1} + n_{2j}$

Results: adaptive criterion eliminates merging error

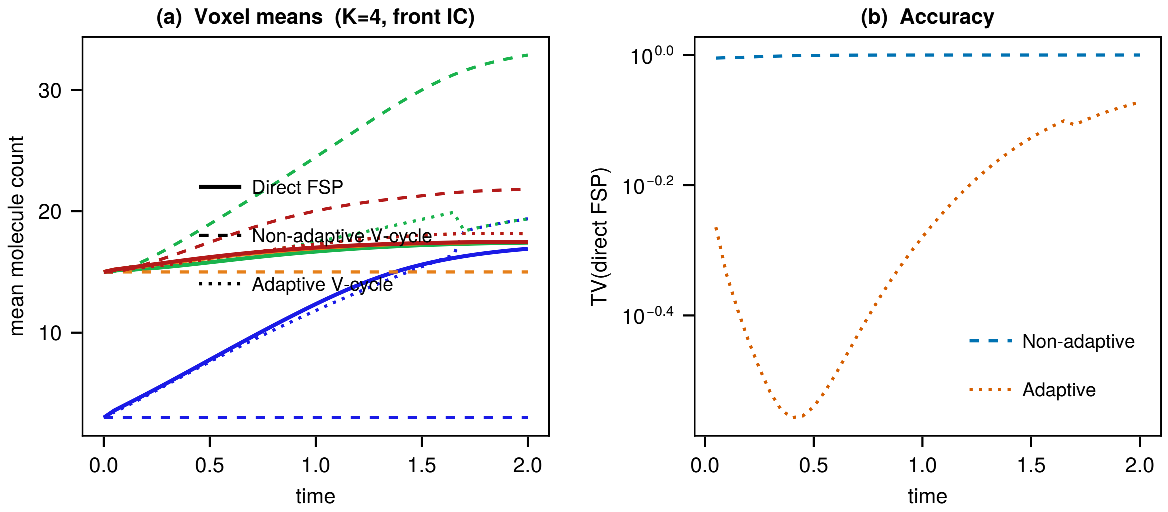


Total variation distance from the reference (full FSP).

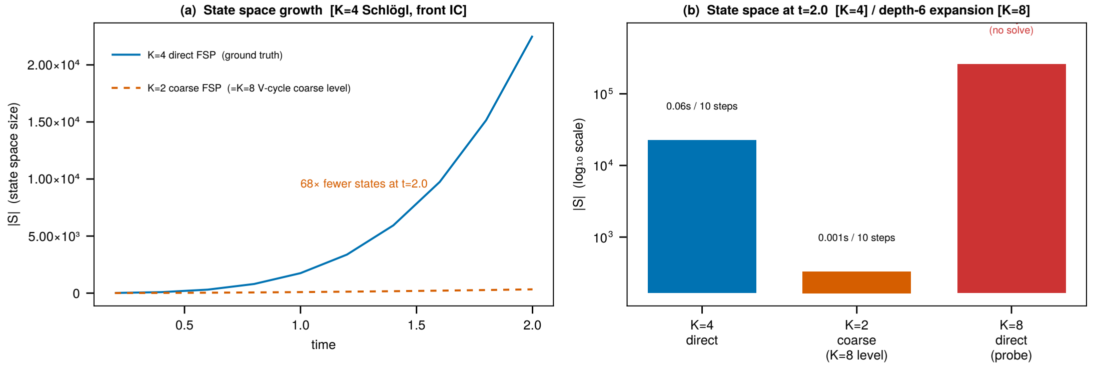
(a) Birth-death: merging from $t{=}0$ gives large transient TV. Adaptive (merge only away from $\text{argmax}_j\ \alpha_j$): TV $\to 0$ after equilibration.

(b) Schlögl front: naïve merging gives TV ≈ 1 throughout (interface orientation lost). Adaptive: TV ≈ 0 throughout.

Results: adaptive mixed-resolution is accurate and fast



Voxel means: adaptive mixed-resolution vs. full FSP. Means match; TV error is small but nonzero (see left panel).



State-space size across K. Mixed-resolution is dramatically smaller; full FSP at large K is infeasible. *(Size comparison, not a timing benchmark.)*

Summary

1. Fast diffusion enables state-space compression.

When the joint distribution of two adjacent voxels is near $\mathrm{Bin}(\bar{n}, \frac{1}{2})$ (verified at runtime via α_j), Replacing (n_1, n_2) by $\bar{n}$ cuts the per-group state space from $n_{\max}^{2S}$ to $n_{\max}^{S}$.

2. Bistable fronts violate the Binomial assumption.

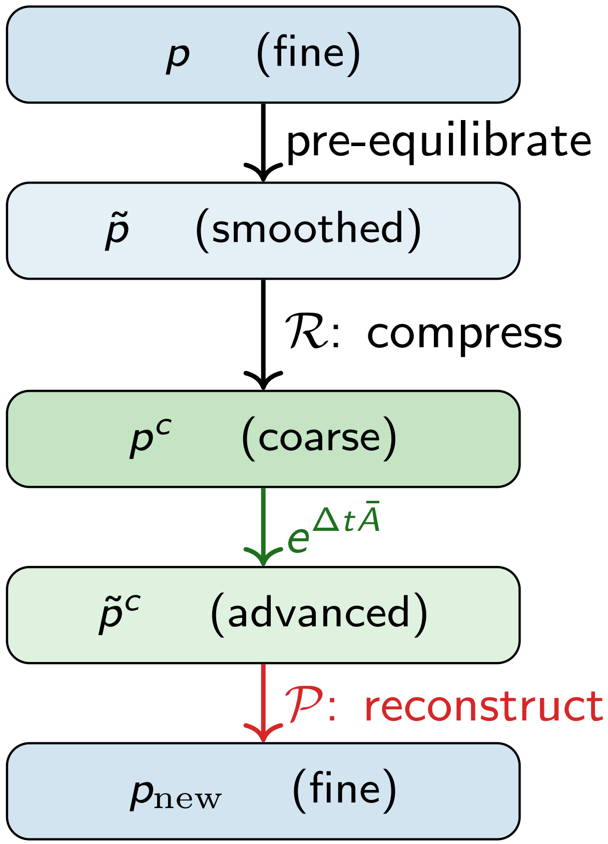
Reactions keep neighboring voxels in opposite stable states, producing a bimodal conditional — not Binomial. The imbalance α_j detects this automatically.

3. Adaptive mixed-resolution FSP.

Maintain a fine window at the front; coarsen the rest. Evolve $\dot{p}^M = A^M p^M$ directly — no reconstruction step. **Accurate and tractable for K where full FSP is not.**

Scope & open questions. Current: 1D, single dominant interface, empirical criterion. Open: rigorous error bound; multi-species; multiple fronts.

Naïve approach: pre-equilibrate $\to$ compress $\to$ coarse solve $\to$ reconstruct.

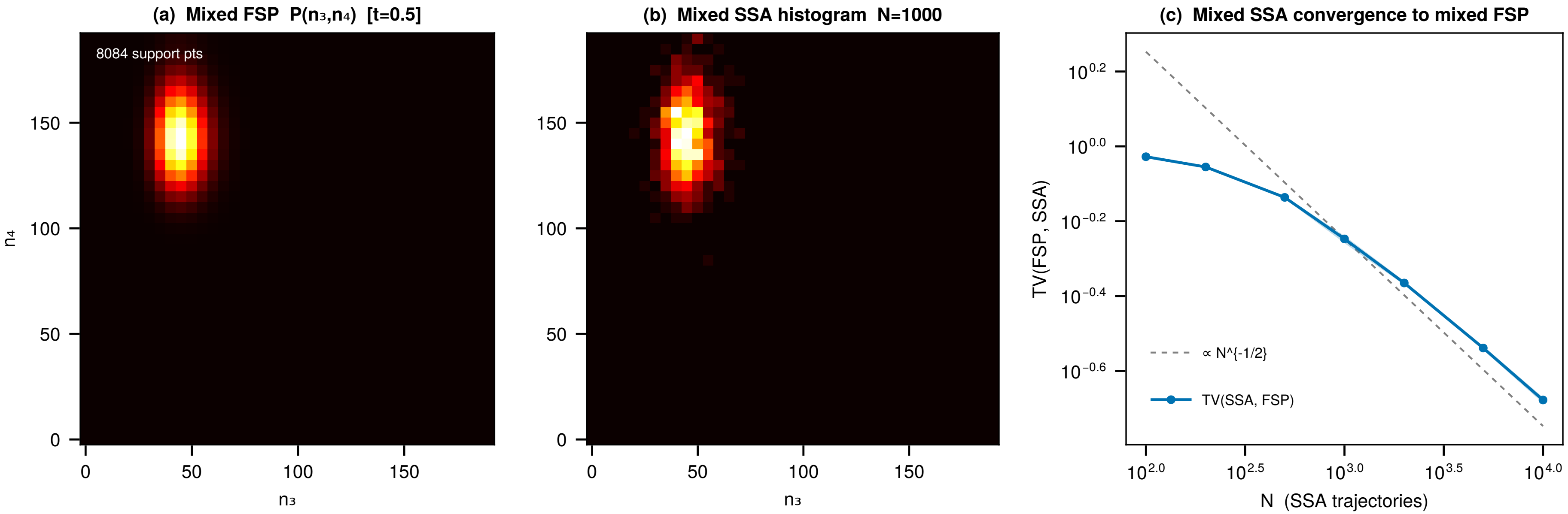


Reconstruction is wrong at the front

Step 4 forces a Binomial split on *every* group of adjacent voxels, including the front where the true distribution is bimodal.

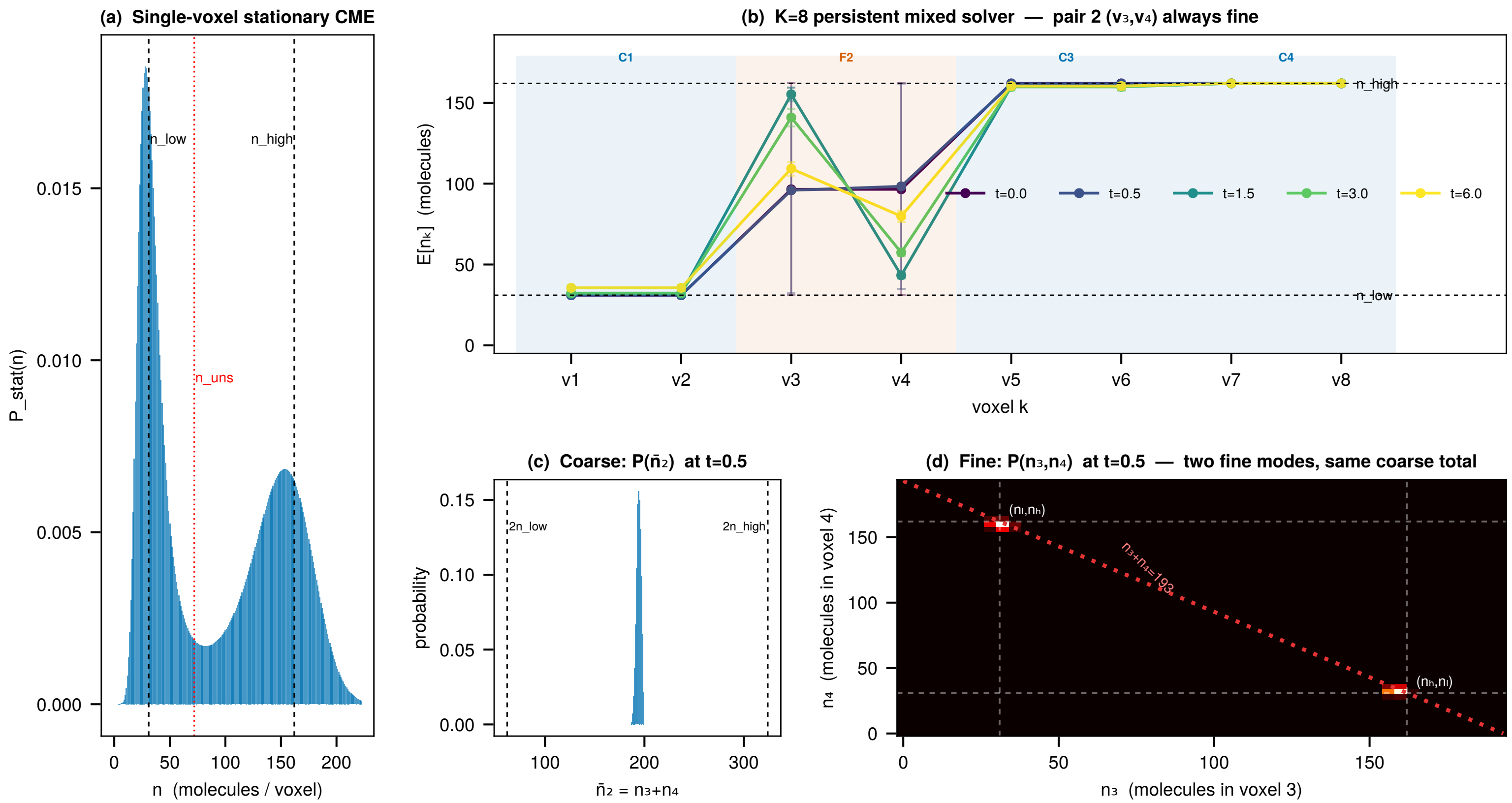
Fix: evolve p^M directly — no reconstruction step. Coarse groups carry the Binomial assumption implicitly in $\bar{A}_j$; the fine group tracks the true within-group distribution exactly.

Appendix: mixed-resolution generator is self-consistent



ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the *same* reduced generator. Means agree to $<1\%$; TV $\sim N^{-1/2}$ (pure sampling noise). This confirms the solver is correct for the reduced model — not that the reduced model is unbiased relative to the full RDME.

Appendix: resolving interface orientation ($K=8$)



IC: interface voxels in equal superposition of $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$. **Coarse** $P(\bar{n}_2)$: single peak — orientation lost. **Fine** $P(n_3, n_4)$: two modes — mixed state retains the within-group structure.

References I

- Erban, Chapman, Maini (2007). Stochastic simulations of reaction-diffusion. *arXiv:0704.1908*.
- Munsky & Khammash (2006). Finite State Projection. *J. Chem. Phys.*, 124:044104.
- Briggs, Henson, McCormick (2000). *A Multigrid Tutorial*, 2nd ed. SIAM.
- Dendukuri, Chandrasekaran, Petzold (2025). Adaptive multigrid FSP for the RDME. *In preparation*.